

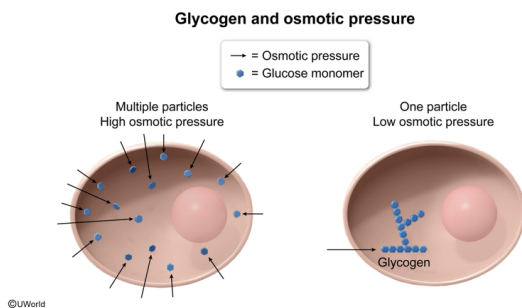
Which of the following statements best explains why animal cells store glucose as glycogen?

- ☐ A. A given volume of water can dissolve a greater mass of glycogen than it can of free glucose. [8%]
- ☐ B. More net ATP is produced per free glucose if it is first converted to glycogen before entering glycolysis. [11%]
- ✓ ☒ C. A glycogen molecule composed of 1,000 glucose units exerts less osmotic pressure than 1,000 free glucose monomers. [49%]
- ☐ D. Glycogen formation requires energy input, making glycogen more thermodynamically stable than glucose. [30%]

Correct

49% answered correctly

Explanation:



The **osmolarity** of a solution describes the **number of solute particles per liter** of that solution. When two solutions with **different osmolarities** are separated by a water-permeable membrane (eg, a cell membrane), **water diffuses across that membrane** from the **lower- to the higher-osmolarity** solution. This behavior—called **osmosis**—allows both sides of the membrane to approach **equal osmolarity**. The tendency of a solution to take in and retain water molecules **due to osmolarity** is quantified by its **osmotic pressure**.

Most organisms use glucose as a primary source of energy. When energy requirements are met, **any excess glucose is stored** for later use. However, storing **large amounts of glucose** as monomers results in a **high osmotic pressure**, which can cause the cell to **rupture** as too much water flows in. Because osmolarity and osmotic pressure are related to the number of distinct particles in solution, this problem can be avoided by **linking multiple glucose particles into one large particle** (ie, a **polysaccharide**). In the case of animal cells, the large particle is **glycogen**.

Thousands of glucose molecules can be combined into one glycogen particle. Combining 1,000 glucose molecules into a single glycogen particle results in an osmotic pressure 1,000 times lower than would be achieved if the glucose molecules were stored as individual units.

(Choice A) Glucose is highly soluble in water whereas glycogen has a low solubility. Therefore, a given volume of water can dissolve a greater mass of *glucose* than it can of glycogen.

(Choice B) Converting glucose to glycogen requires **ATP and UTP hydrolysis**. **Regenerating the used UTP** requires additional ATP hydrolysis. Consequently, converting glucose to glycogen and then breaking down glycogen for use in glycolysis results in less net ATP production than if the glucose had entered glycolysis directly. Note that degradation of preexisting glycogen followed by glycolysis produces more ATP than glycolysis alone, but the investment of ATP required to synthesize glycogen from glucose in the first place counteracts this effect.

(Choice D) Thermodynamic stability is achieved by bringing a molecule to a *lower* energy state. The fact that glycogen formation requires energy *input* means that glycogen is *less* thermodynamically stable than glucose.

Educational objective:

Animal cells store glucose as glycogen, which reduces the osmotic pressure exerted on the cell membrane.

Time spent:0 sec

Subject:
Biochemistry

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System:
Carbohydrates, Nucleotides, and Lipids